# Rodrigo Andrés Norabuena Mascaraqui

ECONOMIST | ANALYTICS

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## PROFILE

PUCP economist trained in data science, with experience building end-to-end analytical solutions. My training gives me the judgment to frame the question well and read results with business sense. Practice gave me the tools to integrate data, build models and turn the analysis into products people use every day. What I value most is closing the loop: going from the problem to a clear recommendation.

## EXPERIENCE

**Research Analyst Intern — CENTRUM PUCP** Mar 2025 — present

I led the evolution of the school's research data ecosystem, from manual reports to an analytics intelligence system now in internal use. I worked closely with the head of research management, who brought the business context; the design and development of the data solution were my responsibility.

- Automated in Python a manual macro-and-spreadsheet process that fed SQL Server and the leadership's Power BI dashboards, cutting the refresh cycle from 3 days to 5 minutes.
- Redesigned that system from the ground up around the research strategic plan and leadership's management questions, translating them into indicators of productivity, quality, impact, collaboration and accreditation (AACSB, EQUIS), including a composite faculty evaluation score and a researcher-inactivity alert model.
- Built a Medallion-architecture ETL pipeline (Python, pandas, Parquet) integrating Scopus, Web of Science, JCR, SJR and the ROR and OpenAlex APIs, with fuzzy institution matching, data-quality checks, unit tests (Pytest) and loading into a dimensional model on SQL Server.
- Added publication classification to the pipeline under AACSB and EQUIS criteria and 7 research-methodology variables, using language-model APIs in batch and incremental modes.
- Built, with AI coding assistants, a 10-section web dashboard with its own API, institutional authentication and persistent filters, replacing the previous Power BI dashboard.
- Took sole ownership of the area's data ecosystem after a team restructuring, backed by reproducible technical documentation.

## SELECTED PROJECTS

**Mining's footprint in Peru, 2013–2025** 2026

Where did satellite-visible mining appear in Peru, and how much did it grow by district between 2013 and 2025? **727 → 1,985 km²** — visible mining area, ×2.7 in twelve years.

Stack: Python, MapBiomas, NASA Black Marble, Fay-Herriot, Fixed effects, Stata, Streamlit.

**A leakage-free credit risk model** 2026

Can we predict which credit card client will default next month, with validation free of data leakage? **AUC 0.774** — on test data (Gini 0.548 · KS 0.421).

Stack: Python, scikit-learn, imbalanced-learn, XGBoost, LightGBM, SHAP.

**Who does the fire reach? Wildfires in Peru, 2026** 2026

How many people and buildings are within 1, 2, 5 and 10 km of an active fire, and in which districts? **206,740** — people within 5 km of an active fire (October 21 cut-off).

Stack: Python, GeoPandas, DuckDB, GeoParquet, BallTree, deck.gl, GitHub Actions.

## EDUCATION

**B.A. in Social Sciences, major in Economics** — Pontificia Universidad Católica del Perú 2025

Graduated in December 2024; the bachelor's degree was conferred in April 2025.

**Diploma: Data Science for the Social Sciences and Public Management** — Q-LAB PUCP 2026 – 2027

240 hours, in progress.

## SKILLS

**Quantitative analysis:** Applied econometrics, Machine learning, scikit-learn, XGBoost, SHAP, Time series, Spatial analysis (Moran's I, LISA).

**Languages & databases:** Python, pandas, NumPy, SQLAlchemy, GeoPandas, SQL Server, PostgreSQL / PostGIS, R, Stata.

**Data engineering:** ETL / ELT, Medallion architecture, Dimensional modeling, Parquet, Apache Airflow, Apache Spark, Docker, FastAPI, Pytest, Git, GitHub Actions.

**Visualization:** Power BI (DAX, Power Query), Streamlit, Plotly, Advanced Excel (VBA, Power Query).

**Cloud:** Google Cloud Storage, Compute Engine, BigQuery.

**AI:** AI-assisted development (Claude Code, OpenCode), LLM APIs, LangChain.

**Languages:** Spanish (native), Advanced English — C1 comprehension, B2–C1 production (B2 certificate, Idiomas Católica).

## VOLUNTEERING

**Marketing Director — Revista Económica · PUCP** Aug 2024 — Jan 2025

- Led the area and coordinated editorial planning, collaborations and the monthly Macro Report with the other area directors, managing the publication calendar to meet deadlines.
- Defined the content strategy and ran the magazine's social media, guiding decisions with the KPIs I had designed as a member.

**Marketing Team Member — Revista Económica · PUCP** Apr 2024 — Jul 2024

- Designed and tracked KPIs in Power BI, Python and R to evaluate content performance, using a history built through web scraping.

## COURSES & CERTIFICATIONS

**GCI World 2026 · Global Consumer Intelligence** — Matsuo-Iwasawa Laboratory · The University of Tokyo
Sep – Dec 2026 · taught in English

**Diploma: Data Science for the Social Sciences and Public Management** — Q-LAB PUCP 2026 – 2027 · 240 hours

**Q-LAB Winter School · 3rd edition** (scholarship) — PUCP · Q-LAB Aug 2026

**AI Engineer specialized program** — Data Science Research Perú Feb 2026

**Data Engineer Multicloud specialized program** — Data Science Research Perú Oct 2025

**Statistical models with R and Python** — Laboratorio de Datos Sociales Oct 2025

**Q-LAB Winter School · 2nd edition** (scholarship) — PUCP · Q-LAB Aug 2025

**Impact evaluation of programs and public policies** — Proimpacto · Escuela de Economía y Políticas Públicas
Aug 2024

## PARTICIPATION & ACHIEVEMENTS

- **Participant · official IBM certificate** — IBM Quantum · Qiskit Fall Fest 2026 (2026). Weeks of quantum computing training and a team hackathon. Our project, AeroQ Optimizer, frames airport gate assignment as a quadratic assignment problem and compares the classical optimum with QAOA.
- **Finalist** — OEFA Innova 2025 Datathon (2025). As a team, we built a machine learning model with environmental and epidemiological variables for the early detection of dengue outbreaks.
- **Participant · 21 hours** — 1st SUNASS Datathon 2025 (2025). Data analysis to improve drinking water and sanitation services.
- **Participant** — "Exprésate Perú con Datos" Datathon 2024 (2024). Analysis of Fondo MIVIVIENDA housing-program beneficiaries with a gender-gap lens, using open data.
- **Medalist · 2019–2021** — PUCP Kung Fu Wushu University Team (2021). University competitions with the PUCP team. The medals come without a diploma; the university issued recognition certificates.
- **Participant** — Innovation Challenge 2020 (2020). Solution proposal for a partner organization's challenge.